

Google Play Store Apps

- All that you need to know about the Android market

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— **Group 8**

DATA MANAGEMENT FOR ANALYTICS(DNSC-6305)



Summary & Expectations

- We wrangled data into a dimensional model using csvkit and sql to answer 3 business questions
- Dataset can be used for analysis and comparison across various mobile application popularity, user satisfaction, pricing strategy, free app, priced app, developer.
- Google Play store is a digital platform that provides reviews, ratings, and other information to help users make informed decisions about the apps they choose to download.

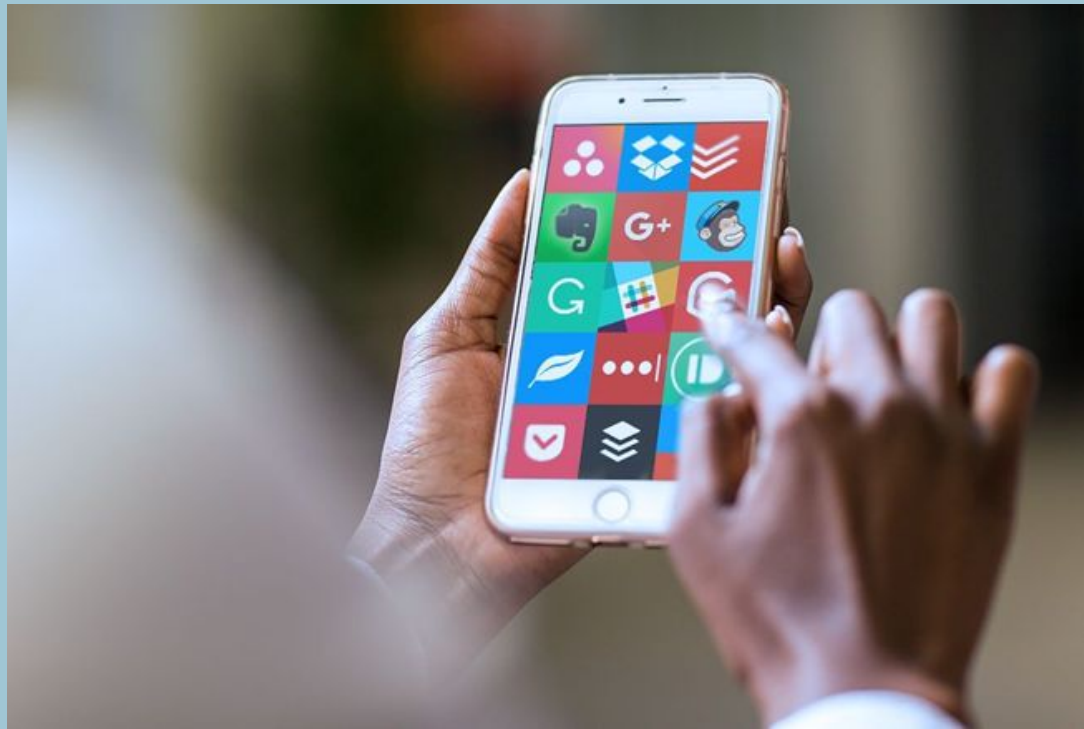




Why this Dataset?

- Fit the requirements for Dimensional Modeling and Analytical Analysis
- Google play store dataset includes a wide range of information and encompasses a comprehensive set of columns pertaining to our dataset.
- World of smart devices with applications. Therefore, the appeal of Google Play Store data resonates with us!

Business Questions



- 01 To analyze the distribution of different categories, applications and rating according to purchase volume.
- 02 Which are the top categories of apps that have an average rating greater than 4.2 ?
- 03 Which developers have the highest number of app installations or top-rated apps during 2020?

Concerns & Modifications

- 01 There is no exact installation number of the apps, so we calculated the average installations.
- 02 There is no continuous date data to check for app downloads in a specific interval of time.
- 03 Some data are missing, so we used csvkit to delete some columns that were not useful, deleted/replaced some NULL data

ASK 2

Data Wrangling

Data cleaning

- Use csvkit cut unnecessary columns
- Replace Null value in Minimum installs, Rating and Rating count to 0/0.0
- Delete the Null instance in Released and Lastupdated columns
- Check Duplicate Data
- The final dataset has 19 columns + 1 column(average installs) and 2241889 records

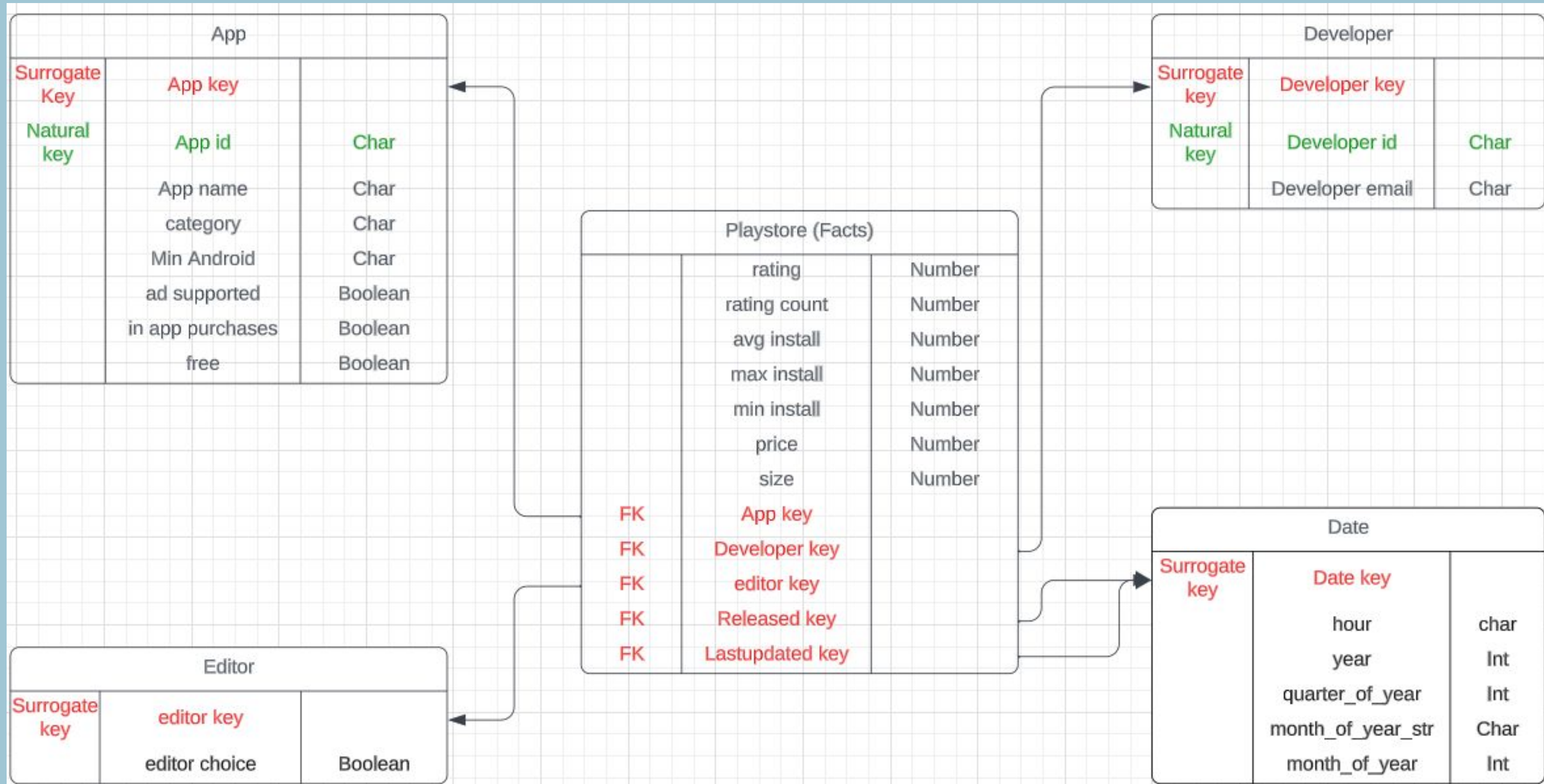
ASK 2

Dimensional Modeling

Google Playstore	
app name	
app id	primary key
category	
rating	
rating count	
install	
max install	
min install	
avg install	
free	
price	
size	
Developer id	
Developer email	
Last updated	
Content rating	
Add supported	
In app purchased	
Editor choice	
released	

ASK 2

Dimensional Modeling

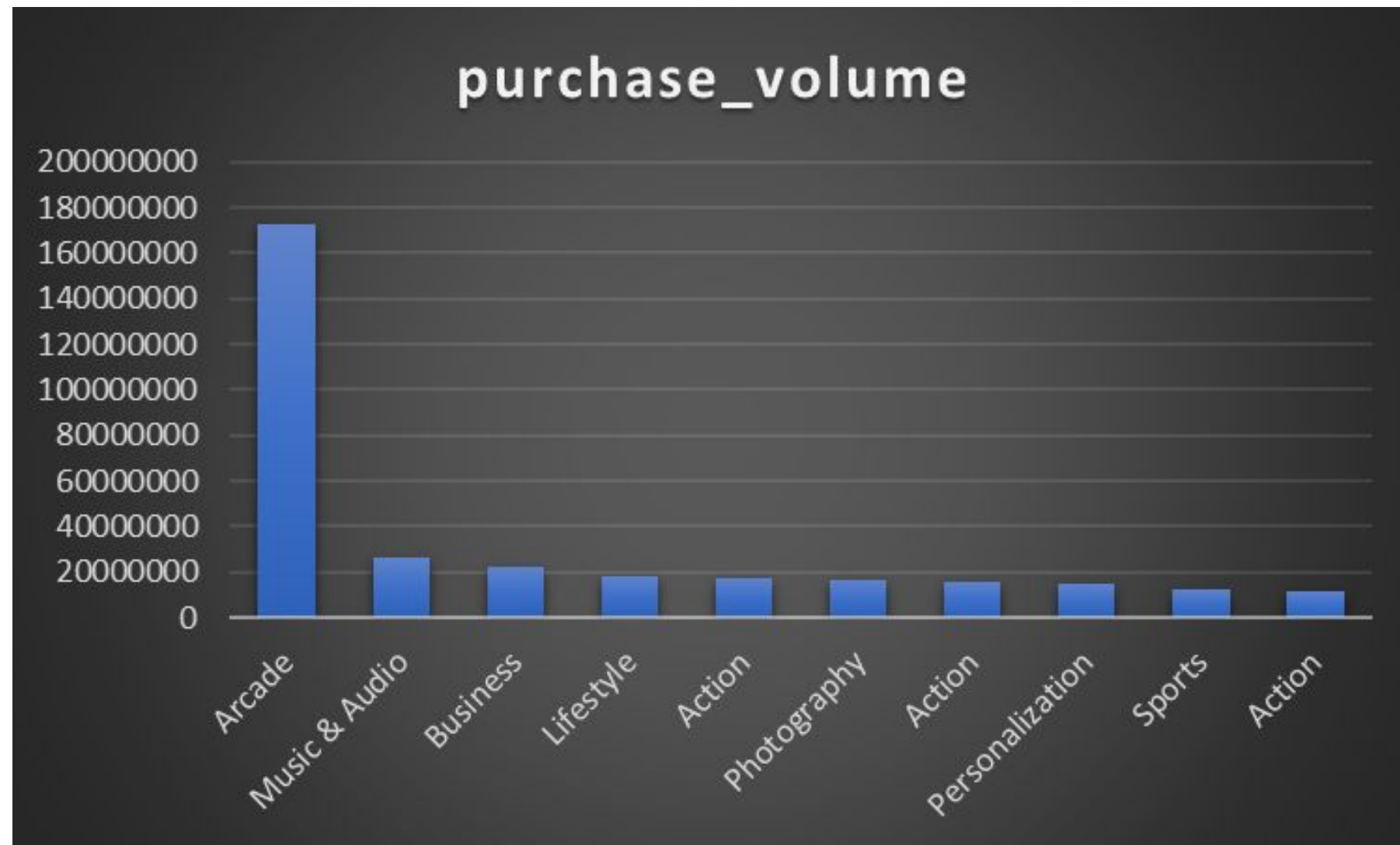


ASK 3 Business Questions



Business Question #1

To analyze the distribution of different categories, applications and rating according to purchase volume.



Finding 1:

- **Arcade** has the highest purchase volume **172956540.89 USD.**
- The second highest purchase category is **Music and Audio** **26152021.14 USD**

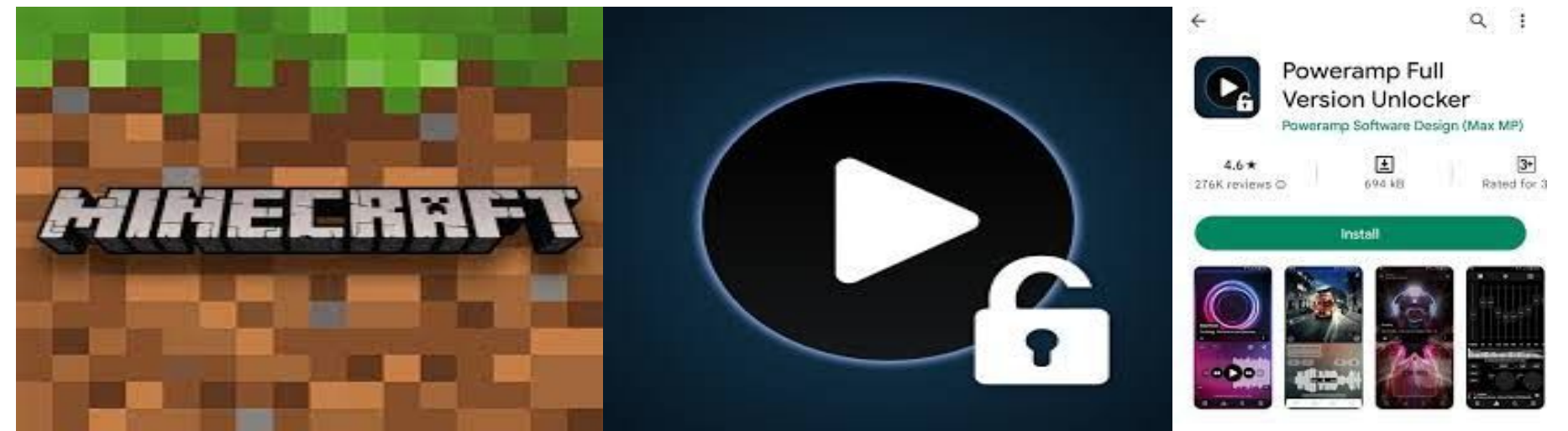
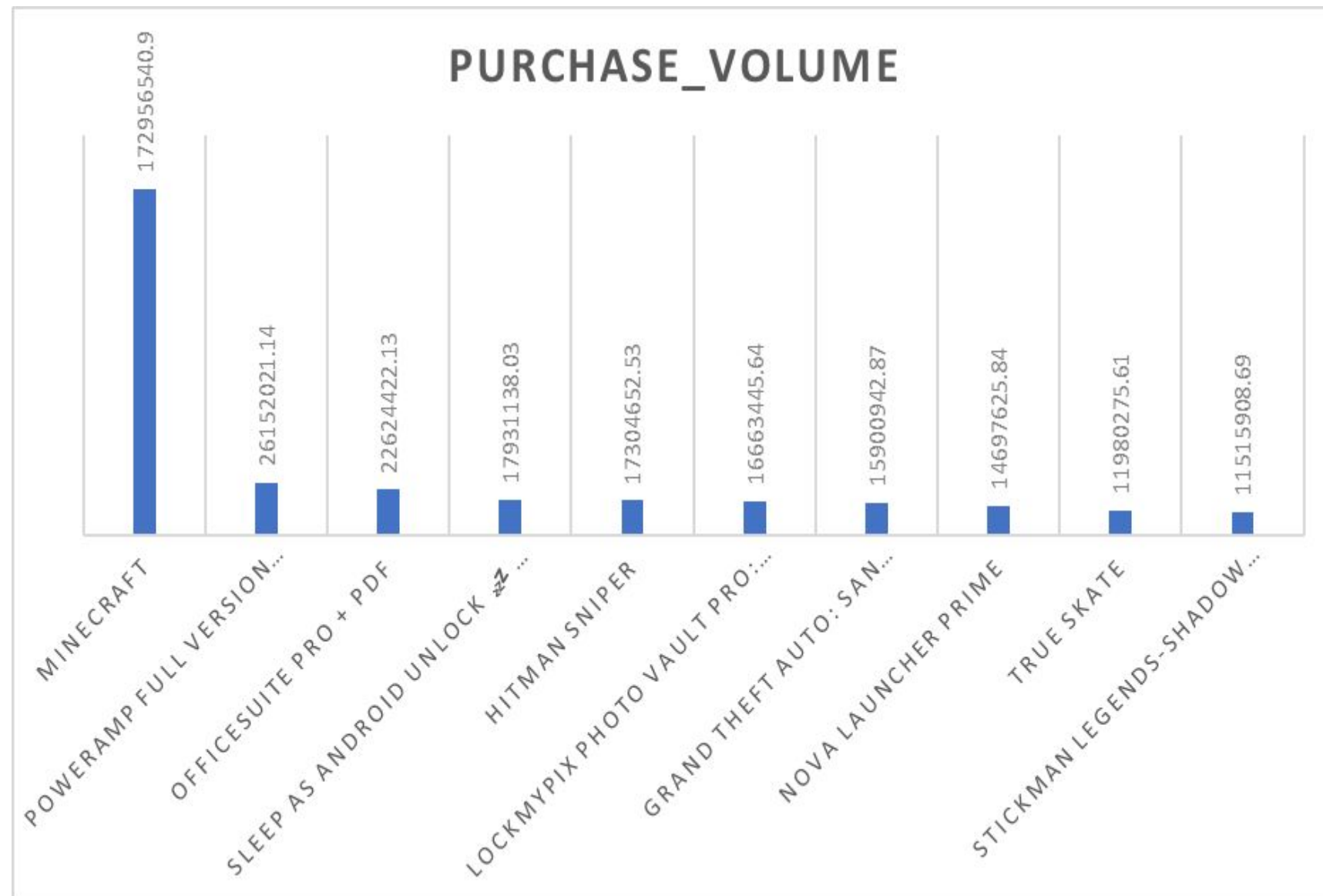


Business Question #1

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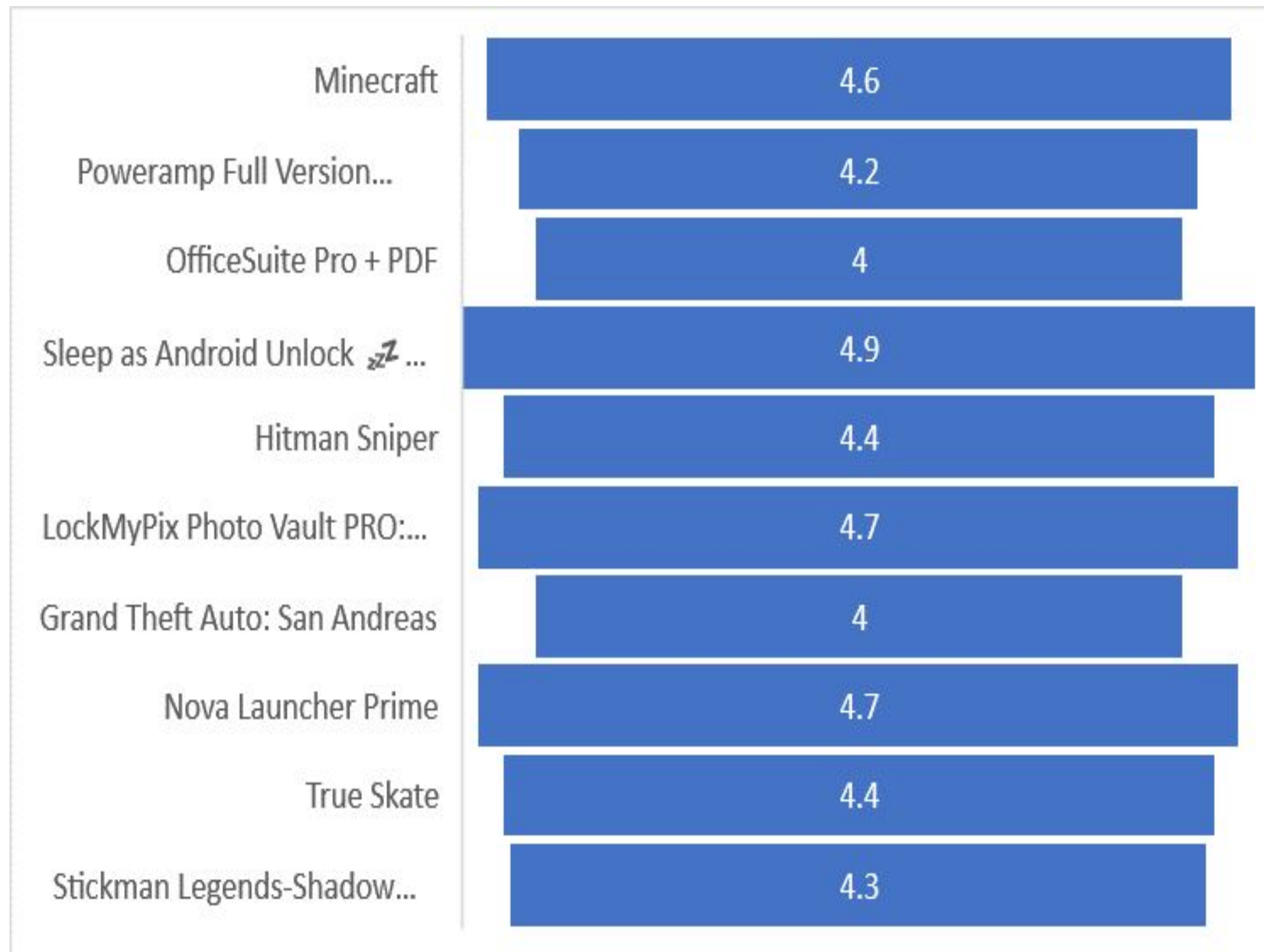
Finding 2:

The most purchased app is **Minecraft** and the second most purchased app is **poweramp full version** thus the startups can focus on similar app to earn money



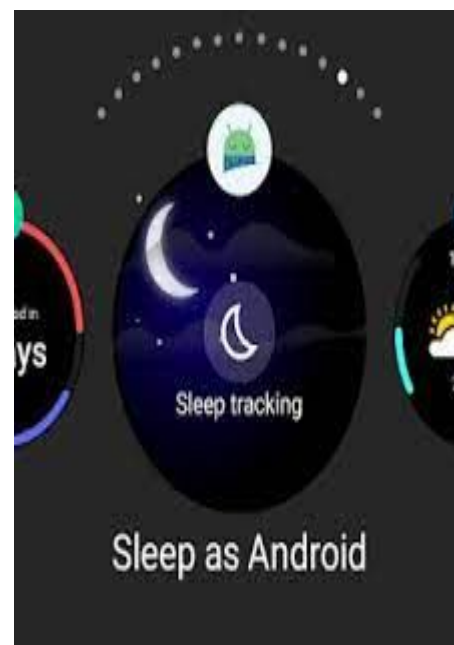
Business Question #1

To analyze the distribution of different categories, applications and rating according to purchase volume.



Finding 3:

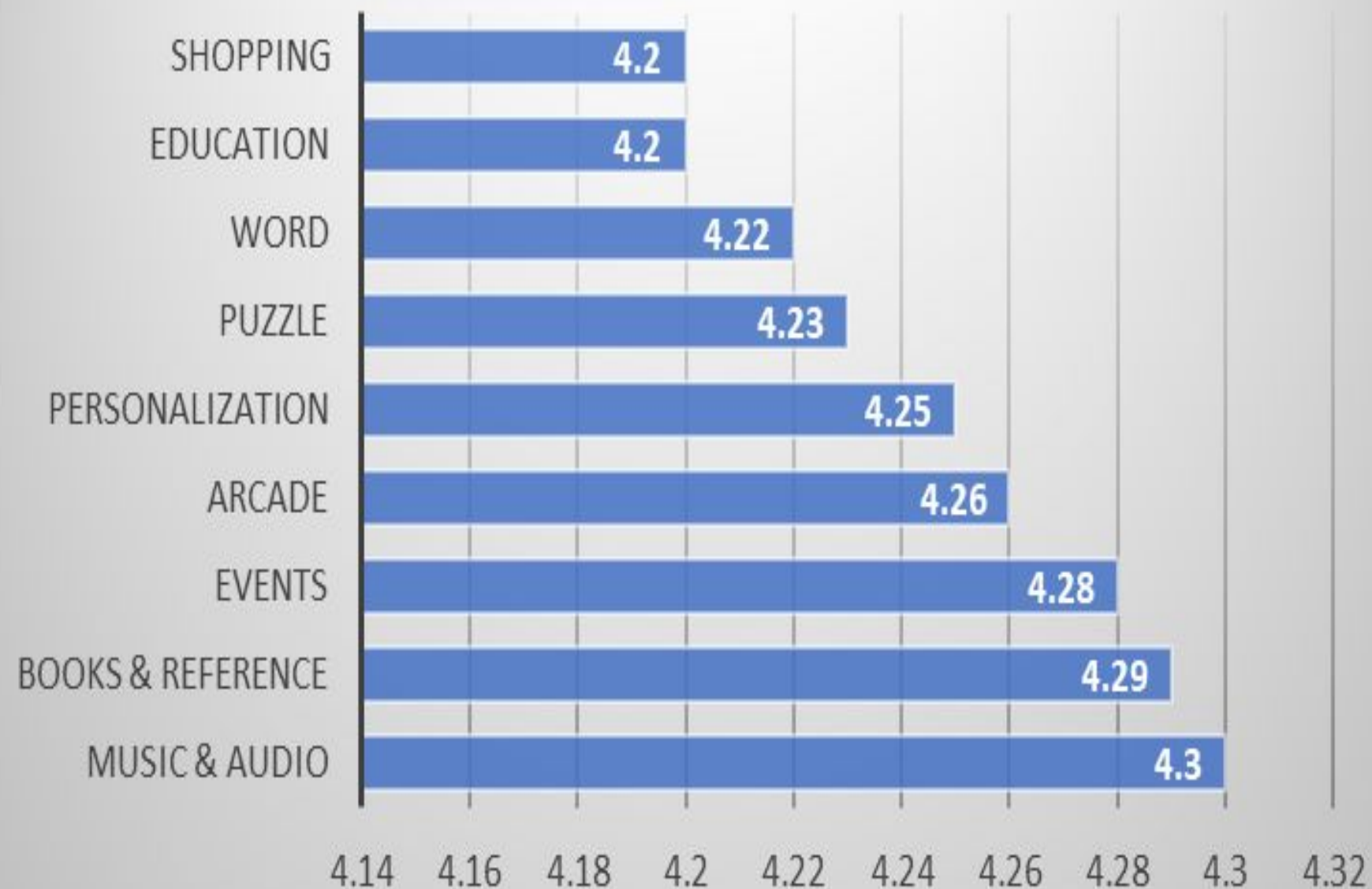
- Check whether people really like the top-selling app, thus incorporating the rating of the top selling app.
- The fact that the highest-purchased app **Minecraft**, doesn't have the highest rating can indeed be surprising, highest rated app is **Sleep as Android**, it highlights that user preferences can vary based on different factors.



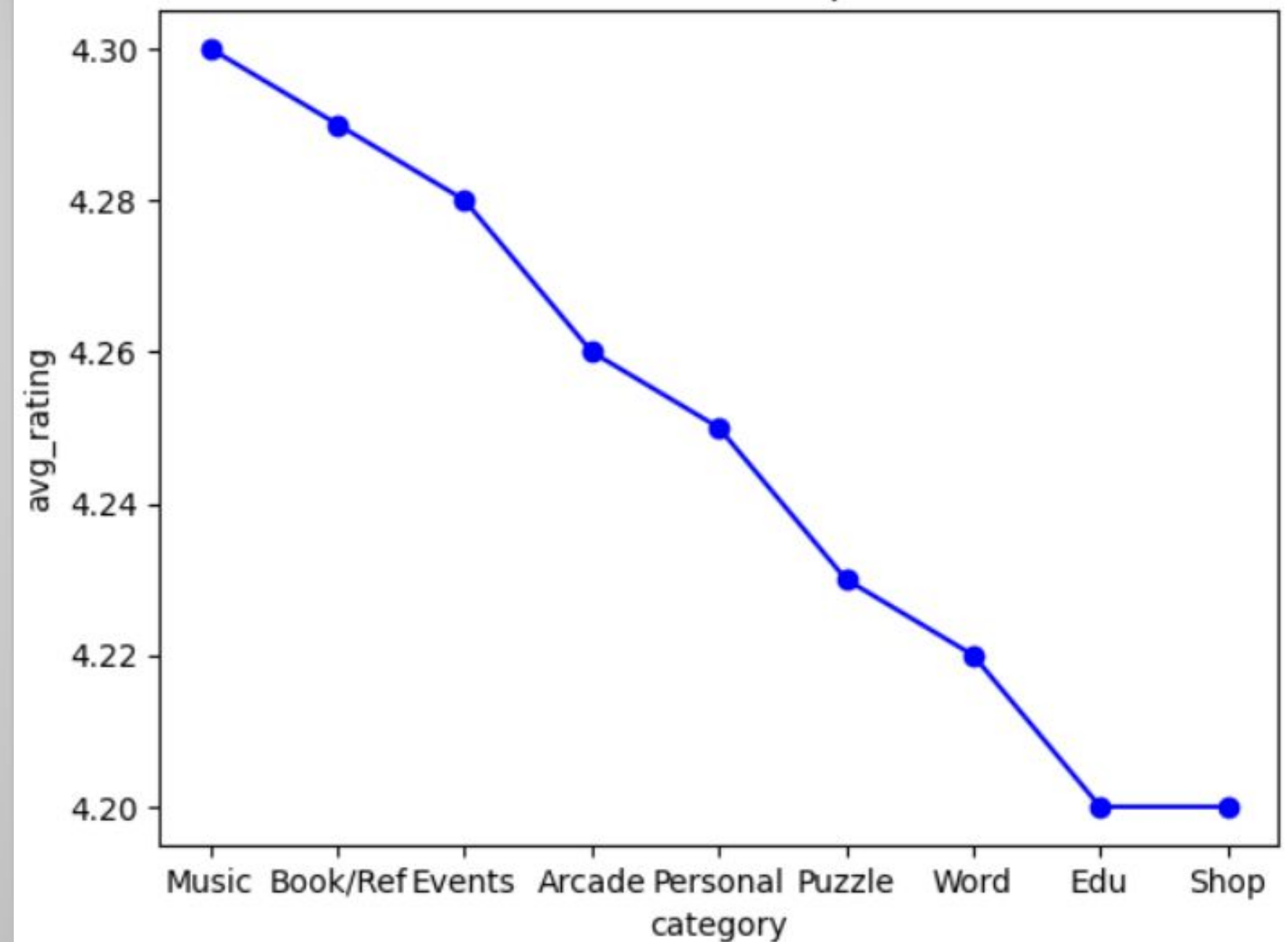
Business Question #2

Identify the top categories of apps that have an average rating greater than 4.2.

Average Ratings for ALL Categories



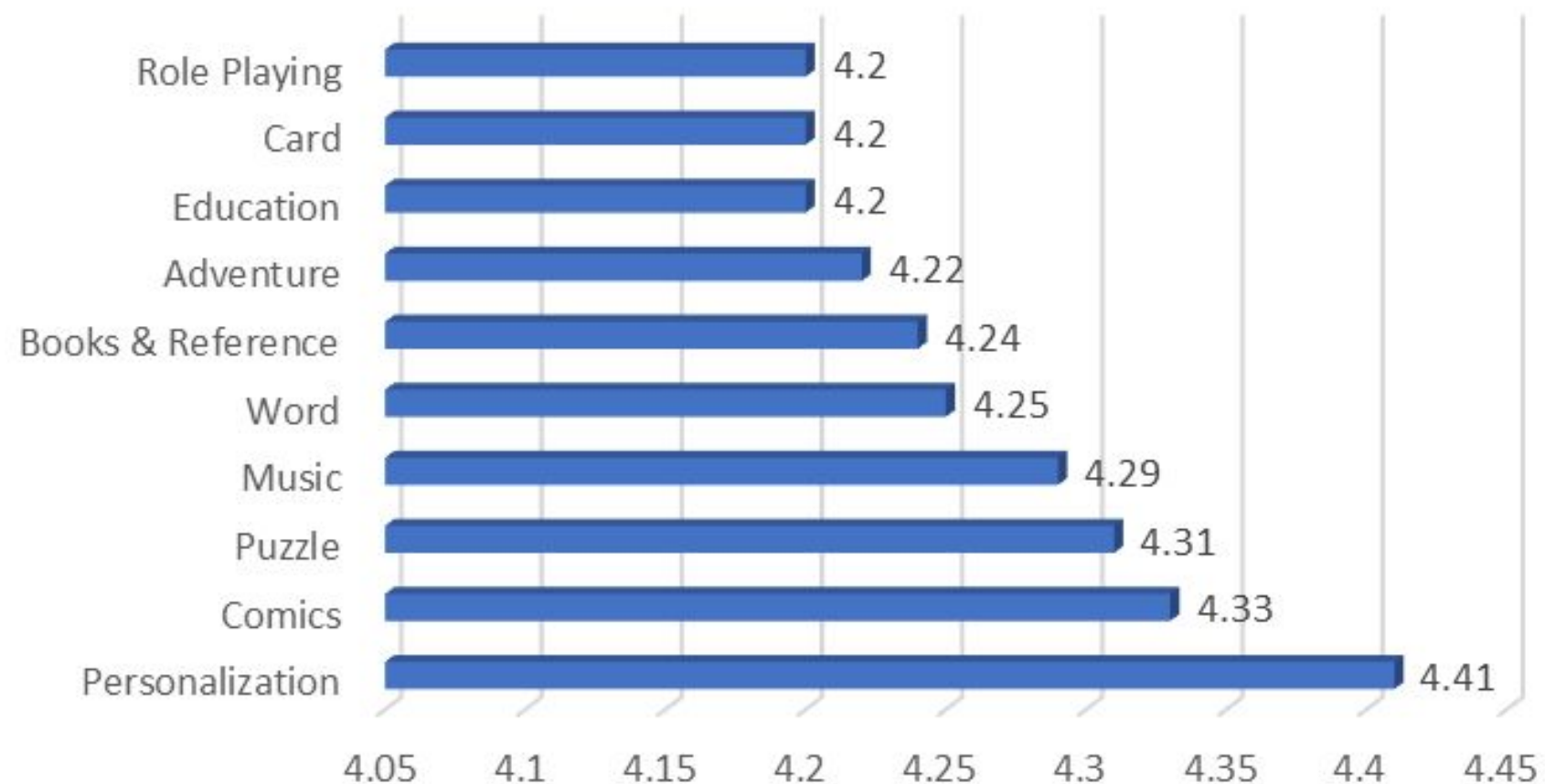
Line Plot Example



Business Question #2

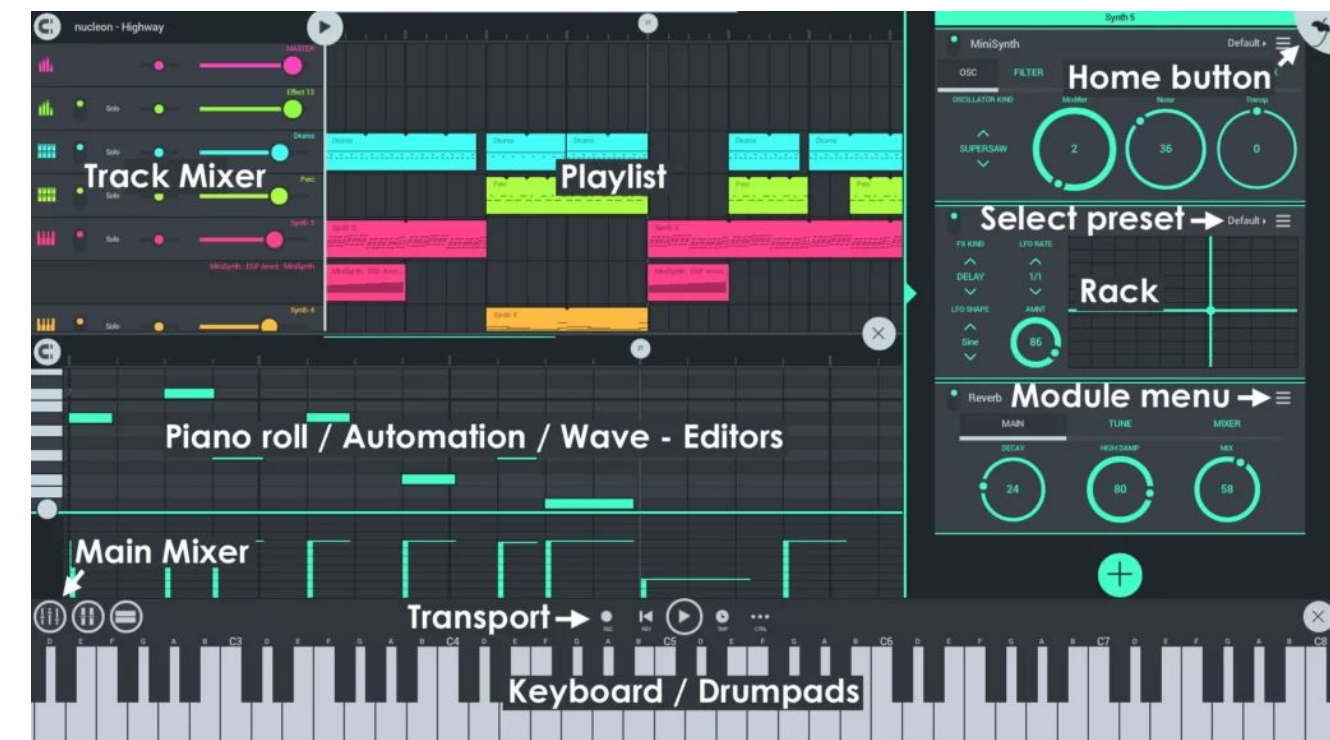
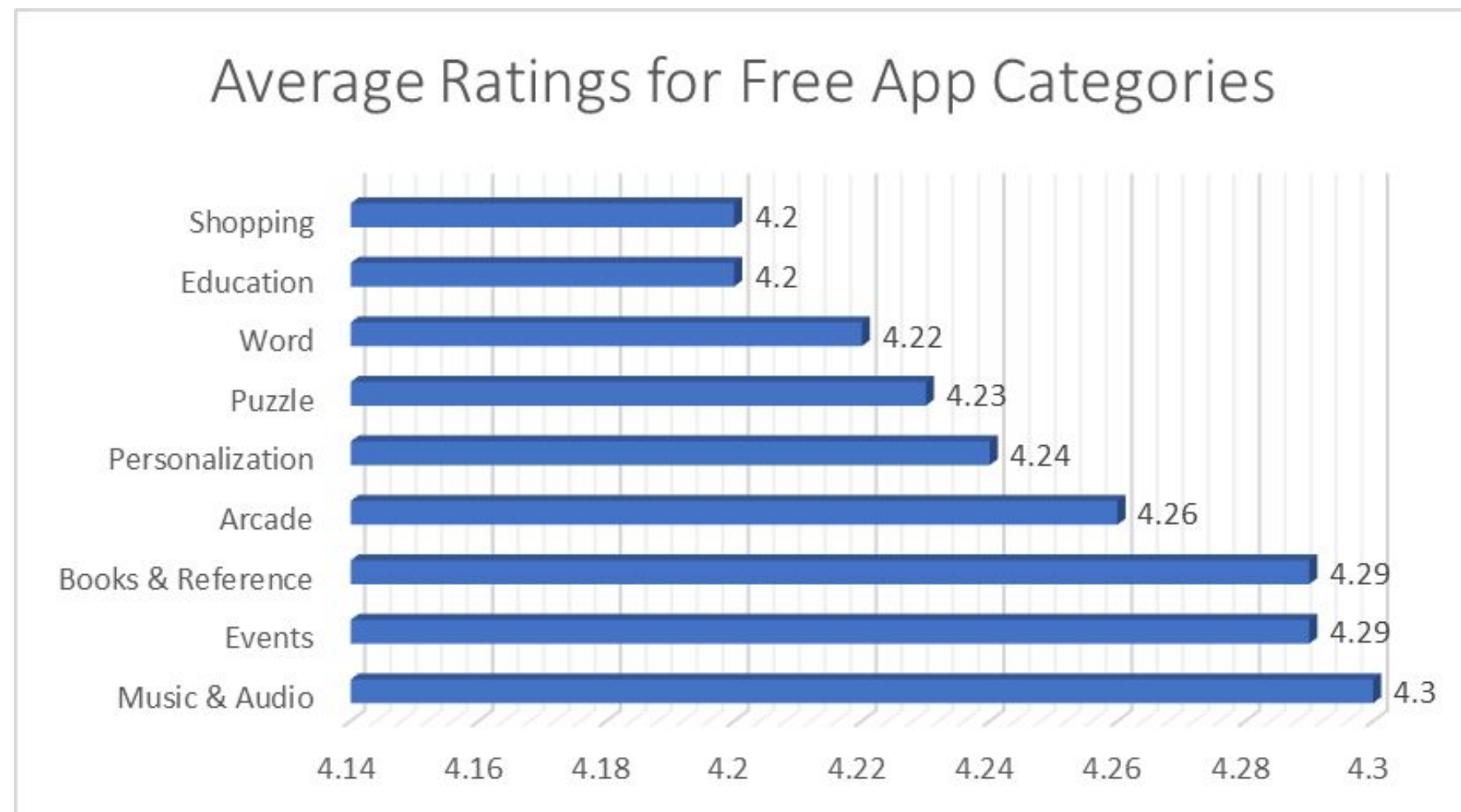
Compare the average ratings between paid and free app within those top categories

Average Ratings for Paid App Categories



Business Question #2

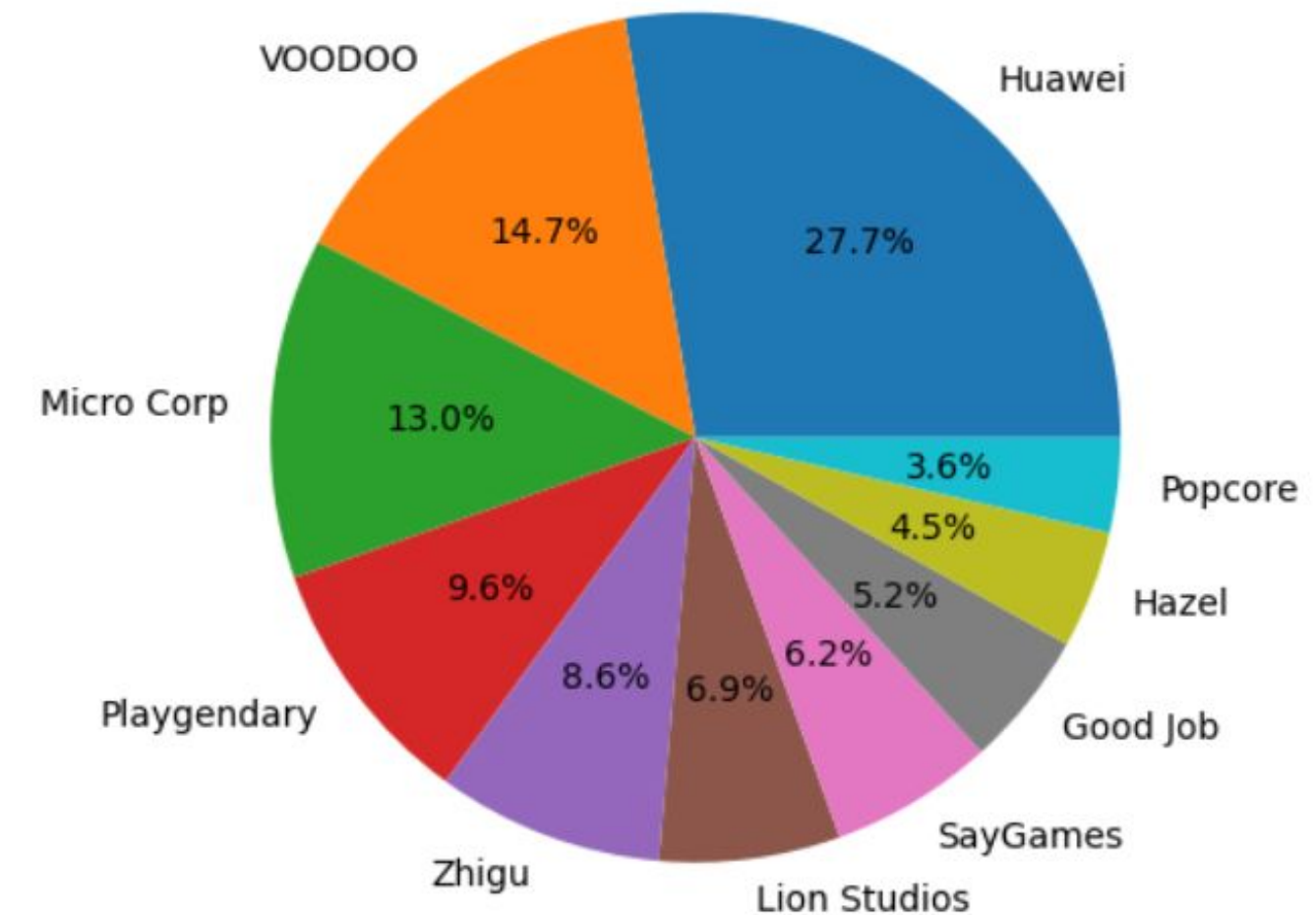
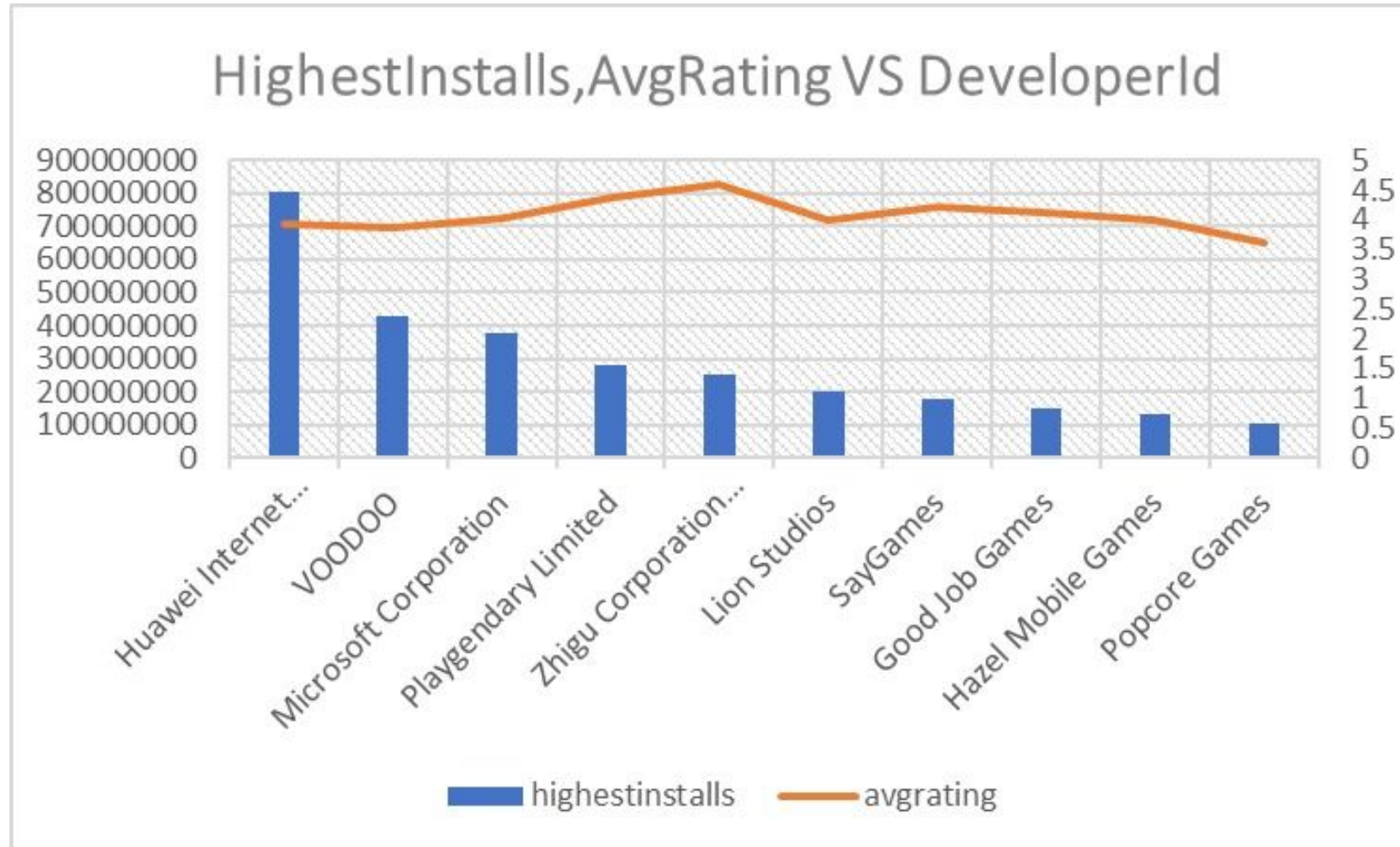
Compare the average ratings between paid and free app within those top categories



Business Question #3

Quarter 1

Which developers have the highest number of app installations or top-rated apps in each quarter of 2020 (covid time)

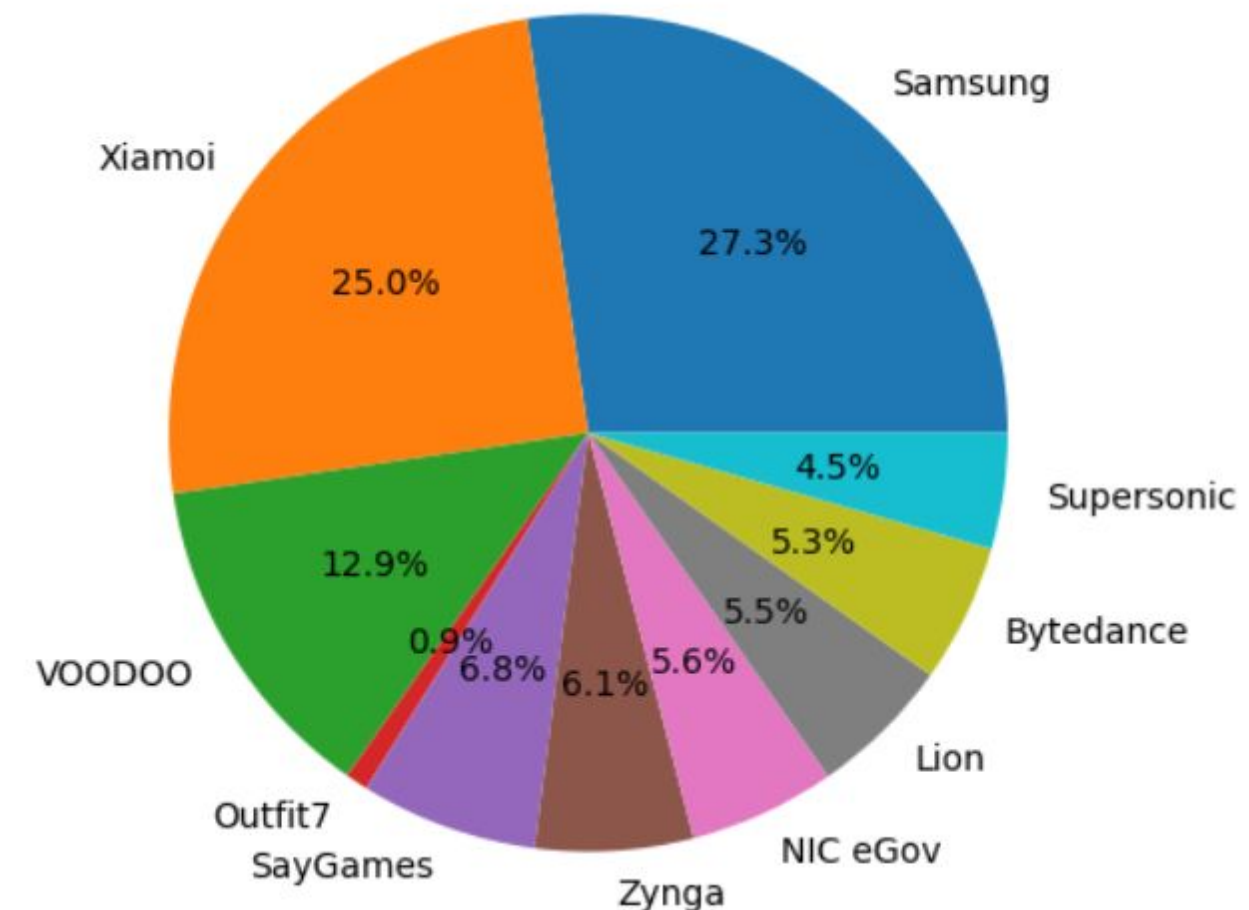
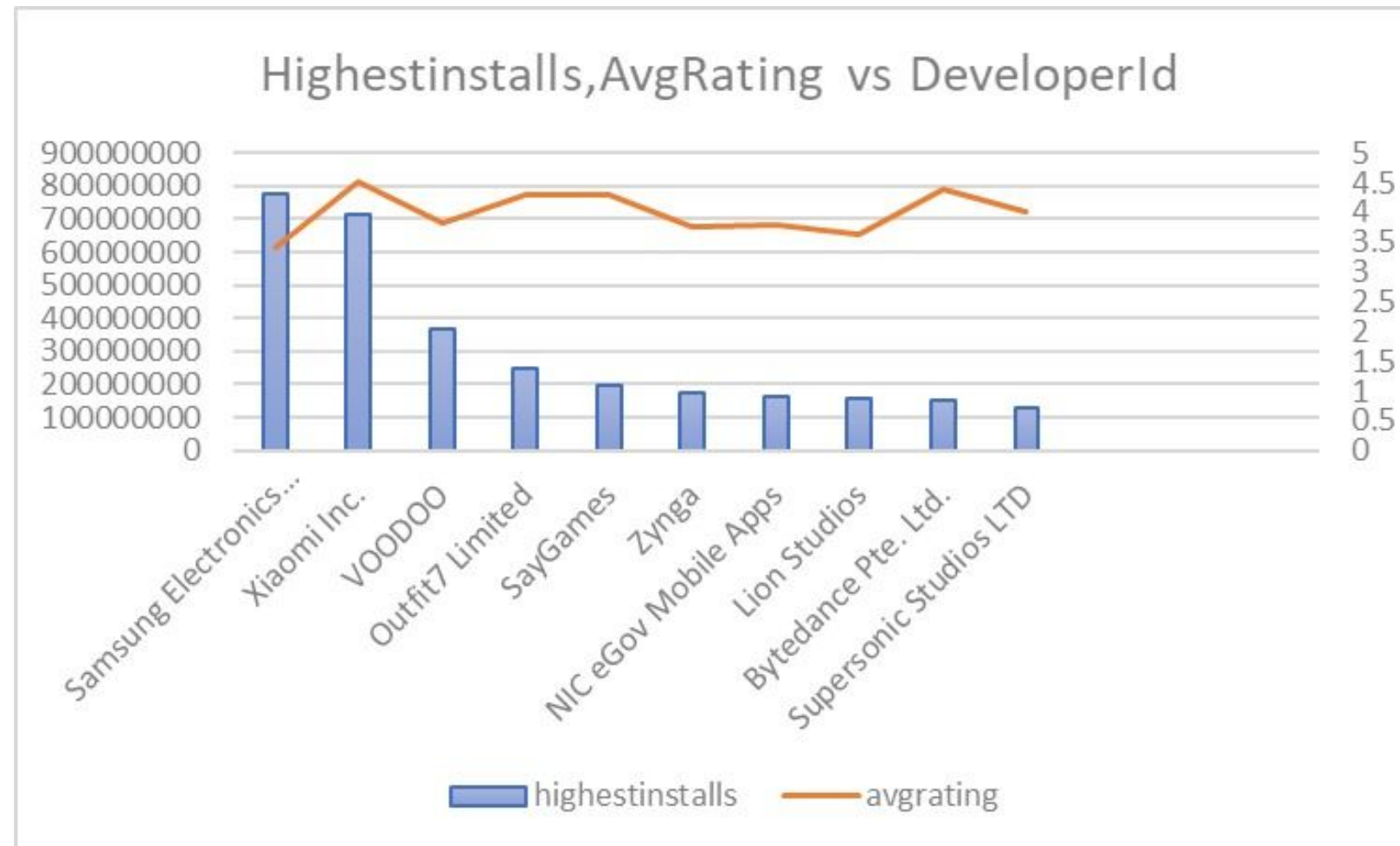


- **Huawei Internet Services** has Highest installations **803,441,420**, Developer has developed **3** Apps, Average Rating **3.93**
- **Voodoo** has **11** Apps, Installations **427,914,673**, Average Rating **3.87**
- **MS corporation** has **3** Apps, Installations **376,282,975**, Average Rating **4.03**

Business Question #3

Quarter 2

Which developers have the highest number of app installations or top-rated apps in each quarter of 2020 (covid time)

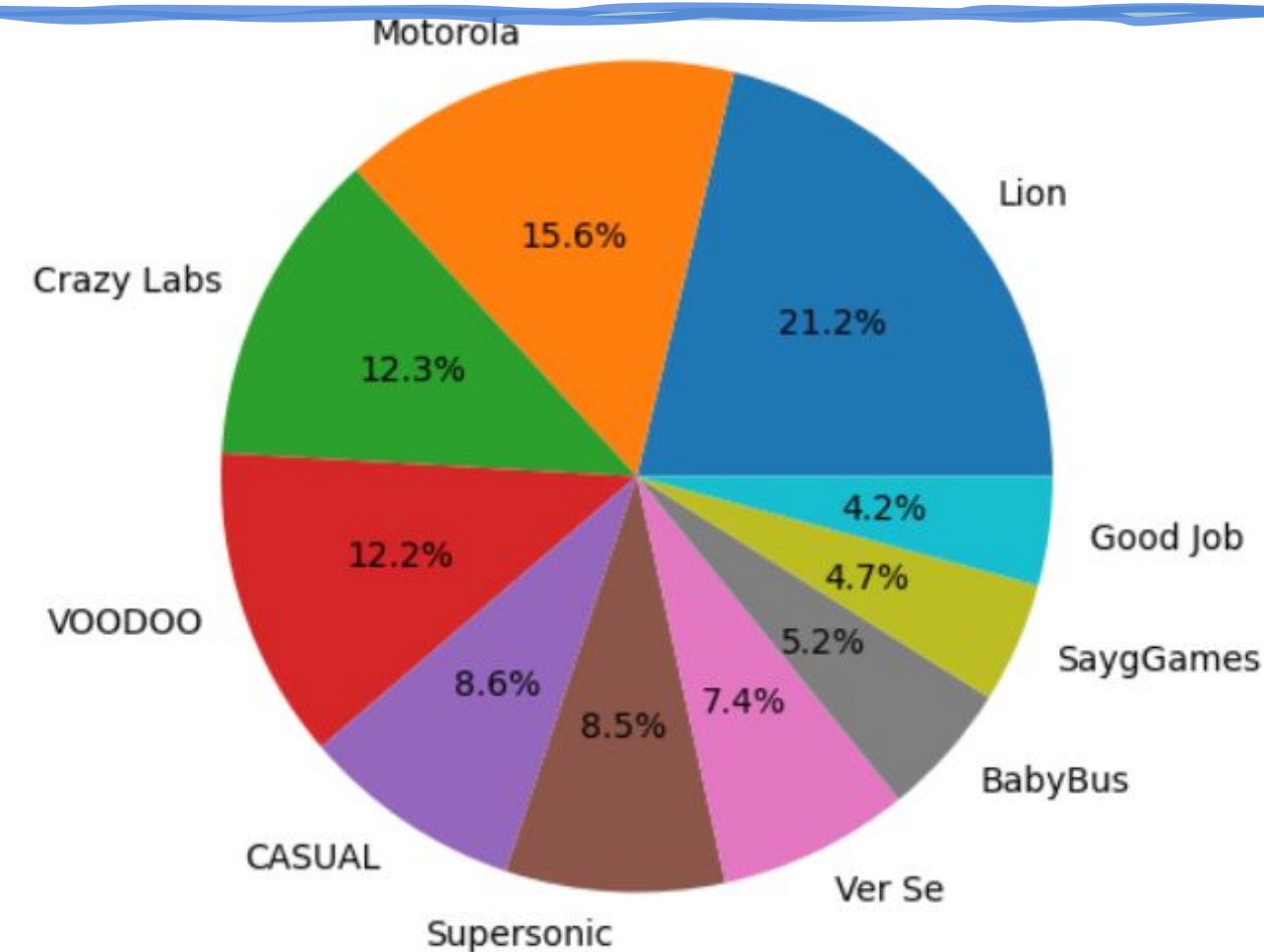
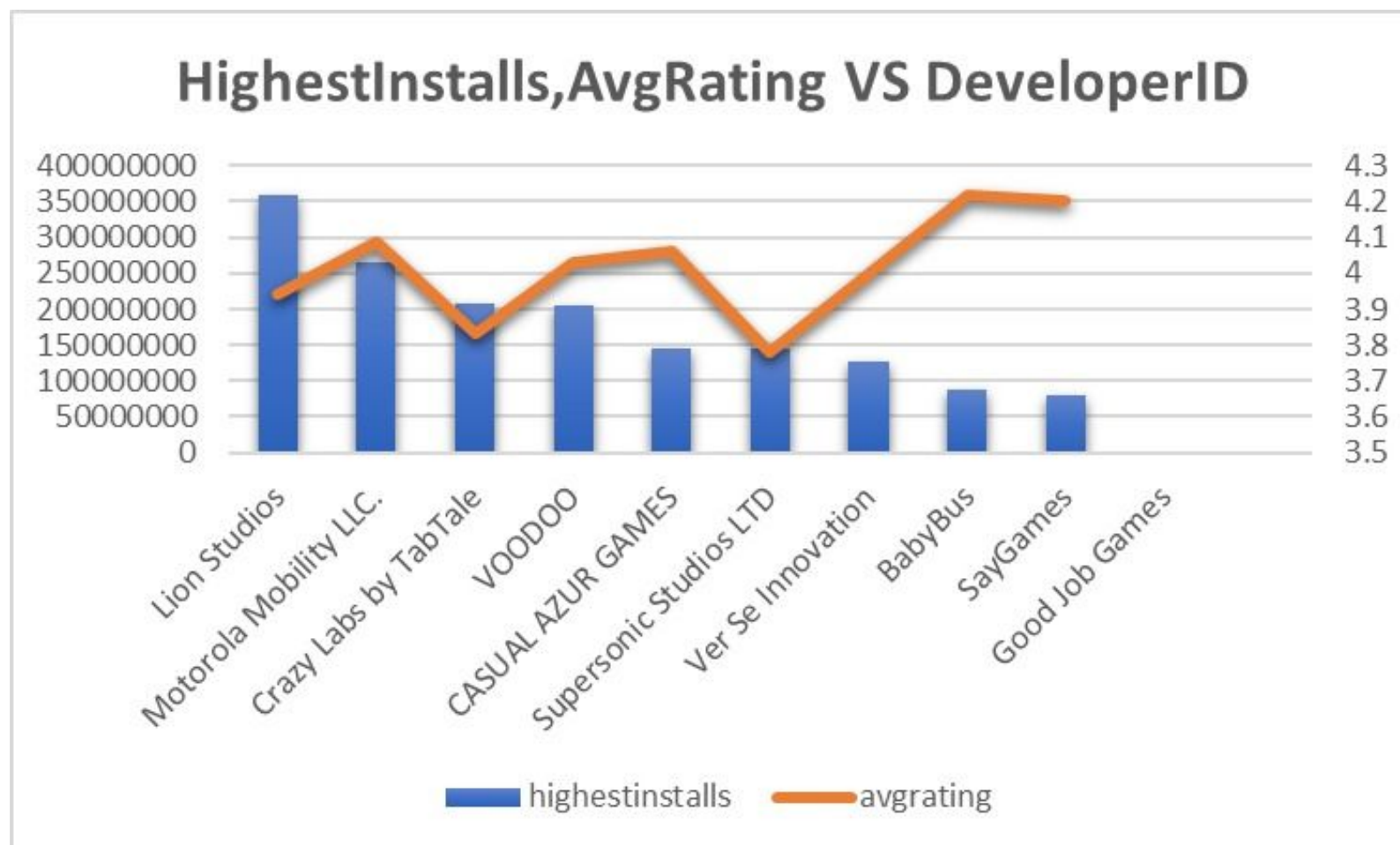


- **SAMSUNG Electronics Co., Ltd** has Highest installations **778,618,060**, Developer has developed **3** Apps, Average Rating **3.40**, Rating count **802,834**
- **Xiaomi Inc** has **1** App, Installations **711,617,649**, Average Rating **4.5**
- **Bytedance Pte. Ltd** has **1** App, Installations **152,325,724**, Average Rating **4.40**

Business Question #3

Quarter 3

Which developers have the highest number of app installations or top-rated apps in each quarter of 2020 (covid time)

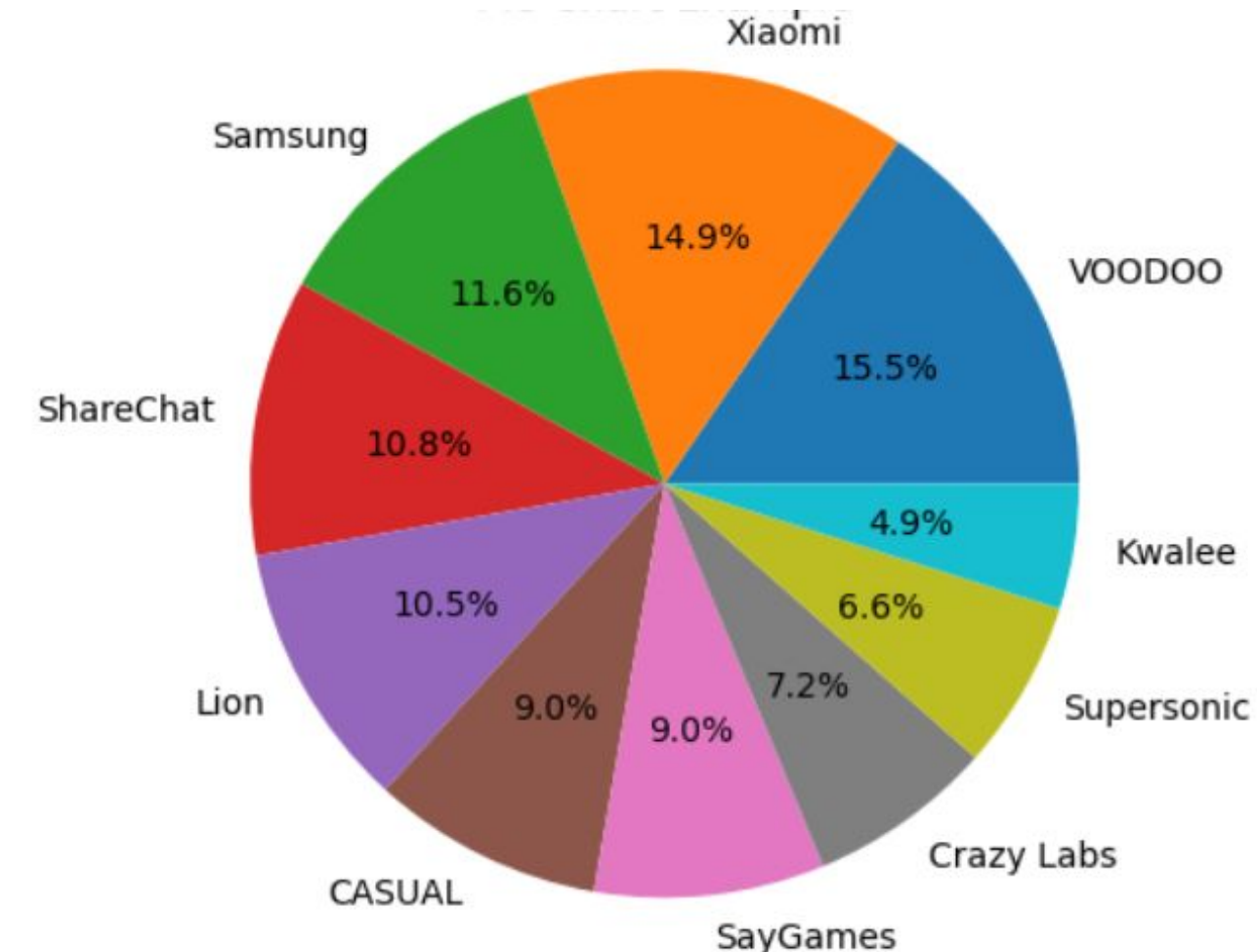
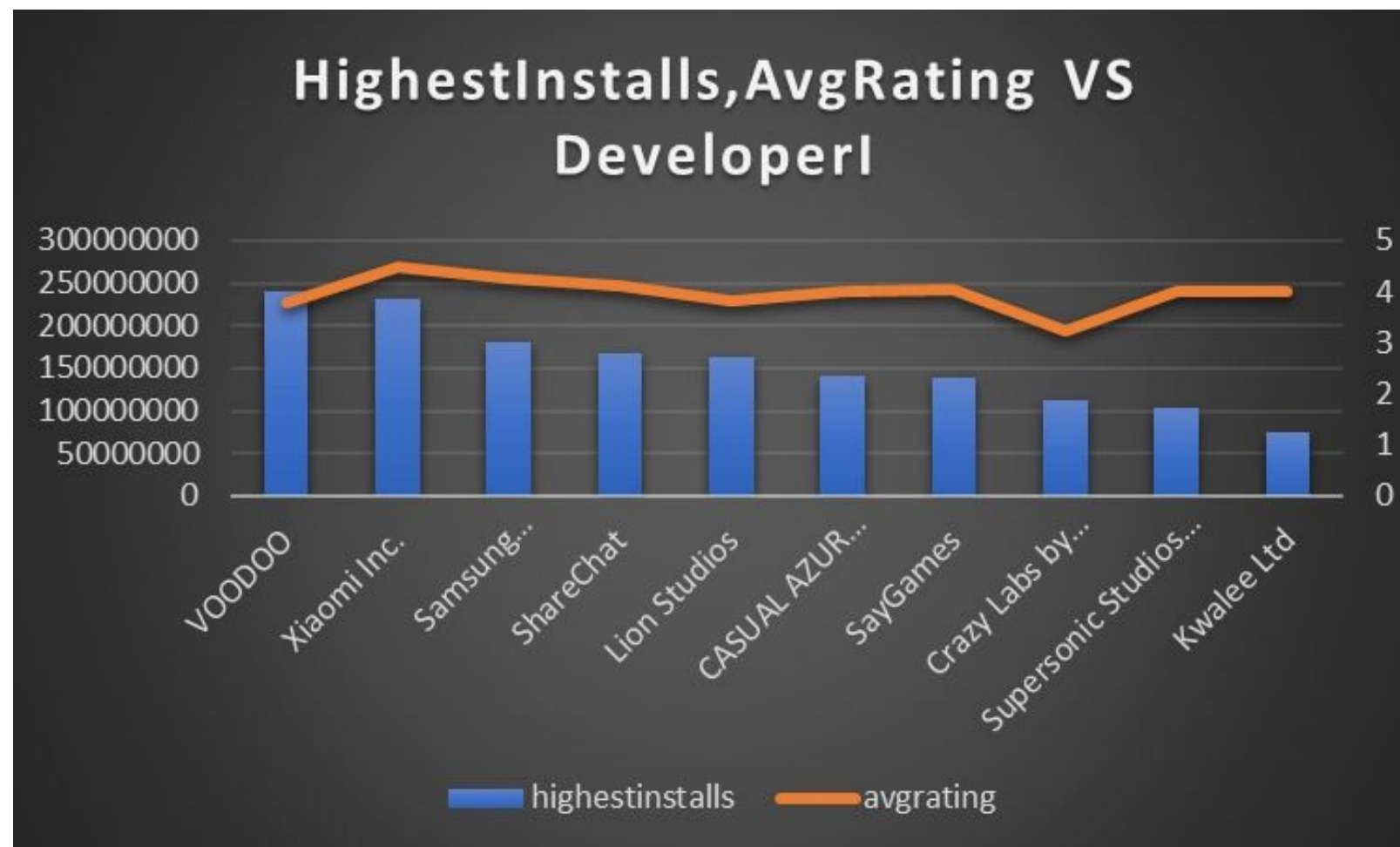


- **Lion Studio** has Highest installations **359,826,775**, Developer has developed **24** Apps, Average Rating is **3.94**, Rating count **280,278**
- **Motorola Mobility LLC.** has **7** Apps, Installations **264,077,130**, Average Rating **4.09**
- **Bytedance Pte. Ltd** has **1** App, Installations **152,325,724**, Average Rating **4.40**

Business Question #3

Quarter 4

Which developers have the highest number of app installations or top-rated apps in each quarter of 2020 (covid time)



- **VODDOO** has Highest installations **239,565,372**, Developer has developed **10** Apps, Average Rating is **3.79**, Rating count **146,215**
- **Xiaomi Inc.** has **2** Apps, Installations **231,010,187**, Average Rating **4.5**, Rating count **30,028**
- **Samsung Electronics Co., Ltd.** has **2** App, Installations **179,774,314**, Average Rating **4.25**, Rating count **6,023**

Thank You!

Q&A